

A new statistic to capture the level dependence in stock price volatility

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Abstract

In this paper, we propose a new covariance estimator based on daily opening, high, low and closing prices. We prove theoretically that the new estimator is unbiased for a pure random walk and further validate it with simulation studies. However, upon examining empirically four indices namely: NIFTY, S&P500, FTSE100 and DAX over the sample period January 1996 to March 2015, we find that the estimator is upward biased for all the indices under study. This overreaction in stock indices can be attributed to the level dependence in stock indices, something that is not captured by the random walk model. So we explore an alternative to random walk, namely: Constant Elasticity of Variance (CEV) specification. Simulation studies provide evidence that the CEV specification can capture the level dependence that makes the estimator upward biased as seen in the data. Therefore, through this specification exercise, we can see that it is possible to isolate the effect of intraday level dependence in stock prices using our estimator.

Keywords: Volatility estimation, Random walk, Extreme values, Covariance, Constant Elasticity of Variance, Level dependence

JEL Classification: C15, C58

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1 Introduction

Volatility estimation has always been an important aspect in pricing and hedging of assets. Starting with the assumption of constant volatility in the Black-Scholes model of option pricing, we have come a long way where we have acknowledged that volatility is not constant, but a stochastic variable. This area of research has attracted a lot of attention, giving rise to studies of alternative estimators of volatility. While the initial focus of most of these studies was on using the daily closing price alone of asset returns for volatility estimation, many studies like [Parkinson \(1980\)](#), [Garman and Klass \(1980\)](#), [Ball and Torous \(1984\)](#), [Alizadeh et al. \(2002\)](#) and [Brandt and Jones \(2006\)](#) have come up with efficient estimators of volatility based on high-low prices. Most of the studies assume that log-prices follow a geometric Brownian motion with or without a drift term.

Such a range-based estimation for volatility using opening, high, low and closing prices of assets have been further extended by many authors like [Rogers and Satchell \(1991\)](#) and [Kunitomo \(1992\)](#), [Magdon Ismail and Atiya \(2003\)](#), [Yang and Zhang \(2000\)](#). [Shu and Zhang \(2006\)](#) have compared the relative performance of the four range based volatility estimators including Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang estimators for the S&P500 index. They find that the estimators are efficient only when the assumption of a geometric Brownian motion for the price of the underlying asset is maintained. From the literature we can see that there have been very few studies that aimed to find a covariance estimator based on intraday opening, high, low and closing prices that are unbiased. This gap motivated us to propose a new unbiased estimator of covariance.

The covariance estimator based on high and low prices introduced in this paper (as explained later) is closely related to the work done by [Rogers and Zhou \(2008\)](#). They are the pioneers in developing a new method of estimating the correlation of a pair of correlated Brownian motions based on the daily opening, high, low and closing prices of stocks.

However, their estimator suffers from the drawback that it is derived under a Brownian motion assumption. Furthermore, it really does not make sense to look for an unbiased

estimator of the correlation when under the null hypothesis $\rho = +1$ because the estimated $\hat{\rho}$ can never be more than 1. Therefore, to be unbiased, the estimated $\hat{\rho}$ would always have to be 1. Similarly even with $\rho = -1$, the same thing is true because $\hat{\rho}$ cannot be less than -1.

Our contribution in this paper is to propose an unbiased covariance estimator based on the high-low prices, where it needs to be noted that the covariance does not have to lie in any particular range.

Apart from [Rogers and Zhou \(2008\)](#), there are some other papers, such as [Brandt and Diebold \(2006\)](#) and [Brunetti and Lildholdt \(2002\)](#) that look at estimating the covariance of foreign exchange rates. [Rogers and Shepp \(2006\)](#) proposed a method to compute the range correlation from linear combination of the four intraday prices namely: open, high, low and close. However, the method we adopt in this paper is much more general compared to theirs.

Further, these papers provide estimators of the covariance only by making use of the assumption of no arbitrage. To put it differently, the solution they provide is too specific to the assets that are under consideration such as foreign exchange rates. That is to say, there is no statistical advance in these papers that can be universally applied.

In this paper, we propose a new covariance estimator *Cov Ratio* which is more general in nature thus making it universally applicable and we are first interested in proving that the proposed estimator is unbiased under the random walk assumption. We find that the estimator proposed is theoretically unbiased. This result is further strengthened with the help of simulation studies under the random walk assumption. However, when we empirically analyze four major stock indices: NIFTY, S&P500, FTSE100 and DAX, we find that the estimator is generally upward biased for all the indices. We conjecture that there exists evidence of level effects on stock price and volatility where by intraday local volatility of stock price is impacted by intraday level of stock prices that makes the estimator biased, something that is not captured by the random walk model. Therefore we explore an alternative to random walk under which such a bias can arise. So, we go for the next alternative, wherein we introduce the Constant Elasticity of Variance

(CEV) specification to capture the level dependence. Using a modified model of the CEV specification for the purpose of our study, we capture the upward bias seen in the data based on simulation exercise.

The rest of the paper is organized as follows. Section two contains the methodology in which we provide the theoretical proof of the unbiasedness of the proposed estimator under the assumption of a random walk. We also introduce the new covariance estimator *Cov Ratio*. Section three deals with the simulation analysis undertaken. The first part of this section deals with the simulation study for a random walk. In the second part of the section, we introduce the CEV specification. We briefly introduce the model and review the past studies related to CEV. We also carry out simulation studies under the CEV specification. Section four covers the empirical findings. The last section concludes the paper.

2 Methodology

In this section, we derive the mathematical proof to show that the proposed estimator is unbiased for a pure random walk under the duality concept.

Consider an ordinary random walk. Assume x_i are iid following any distribution.

$$S_0 = 0$$

$$S_1 = x_1$$

$$S_2 = x_1 + x_2$$

$$S_n = x_1 + x_2 + x_3 + \dots + x_n$$

$$S_N = x_1 + x_2 + x_3 + \dots + x_N \tag{1}$$

We now define (for the ease of notation):

$$x = S_N$$

where S_N is the terminal value of the random walk path

$$b = M_N$$

where $M_N = \max \{S_n : 0 \leq n \leq N\}$

$$c = m_N$$

where $m_N = \min \{S_n : 0 \leq n \leq N\}$

By the duality of random walk which induces the time reversibility in the process; we can define:

$$S_n^* = S_N - S_{N-n} \quad \forall \quad 0 \leq n \leq N \quad (2)$$

$$\text{When } n = 0; S_n^* = S_N - S_{N-n} \rightarrow S_0^* = 0$$

$$\text{When } n = 1; S_n^* = S_N - S_{N-1} \rightarrow S_1^* = x_N$$

$$\text{When } n = 2; S_n^* = S_N - S_{N-2} \rightarrow S_2^* = x_N + x_{N-1}$$

$$\text{When } n = N; S_n^* = S_N - S_{N-N} \rightarrow S_N^* = x_N + x_{N-1} + \dots + x_2 + x_1 \quad (3)$$

From equations (1) and (3), in terms of **joint distribution**

$$\{S_n : 0 \leq n \leq N\} \stackrel{d}{=} \{S_n^* : 0 \leq n \leq N\} \quad (3.a)$$

Result 2.1: $S_N^* = S_N$

Let us define: $M_N^* = \max \{S_n^* : 0 \leq n \leq N\}$

From **(2)**, we can rewrite:

$$M_N^* = \max \{S_N - S_{N-n} : 0 \leq n \leq N\}$$

$$M_N^* = S_N - \min \{S_{N-n} : 0 \leq n \leq N\}$$

Recall: where $m_N = \min \{S_n : 0 \leq n \leq N\}$

Result 2.2: $M_N^* = S_N - m_N$

(3.a) implies that

$$\{S_N, M_N\} \stackrel{d}{=} \{S_N^*, M_N^*\} \quad (3.b)$$

We also know that $M_N^* \stackrel{d}{=} M_N$

From Results 2.1 and 2.2, in terms of the joint distribution:

$$\{S_N^*, M_N^*\} = \{S_N, S_N - m_N\} \quad (4)$$

LHS product and expectation is: $E[S_N^* M_N^*] = E[S_N M_N]$

RHS product and expectation is: $E[S_N (S_N - m_N)]$

On equating LHS and RHS, we get:

$$E[S_N M_N] = E[S_N (S_N - m_N)]$$

$$E[S_N M_N] = E[S_N^2] - E[S_N m_N]$$

On rearranging terms we get:

$$E[S_N (M_N + m_N)] = E[S_N^2] \quad (5)$$

On subtracting $E[M_N + m_N] E[S_N]$ from both LHS and RHS, equation (5) becomes:

$$E[S_N (M_N + m_N)] - E[M_N + m_N] E[S_N] = E[S_N^2] - E[M_N + m_N] E[S_N] \quad (6)$$

Recall: $E[M_N^*] = E[M_N] = E[S_N - m_N] = E[S_N] - E[m_N]$

Result 2.3: $E[M_N + m_N] = E[S_N]$

Equation (6) can be rewritten as:

LHS: $E[S_N (M_N + m_N)] - E[M_N + m_N] E[S_N] = \text{cov}(M_N + m_N, S_N)$

based on Result 2.3

$$\textbf{RHS: } E[S_N^2] - E[S_N] E[S_N] = E[S_N^2] - (E[S_N])^2 = \text{var}(S_N)$$

Thus we get:

$$\text{cov}(M_N + m_N, S_N) = \text{var}(S_N) \quad (7)$$

Here: $x = S_N$; $b = M_N$; $c = m_N$;

Rewriting in terms of the proposed covariance estimator:

$$\text{Cov Ratio} = \frac{\text{cov}(M_N + m_N, S_N)}{\text{var}(S_N)} \quad (8)$$

In terms of x , b and c , the proposed covariance estimator can be expressed as:

$$\text{Cov Ratio} = \frac{\text{cov}(b + c, x)}{\text{var}(x)} \quad (9)$$

Thus we proved that $\text{Cov Ratio} = 1$ i.e. the proposed covariance estimator is unbiased for a pure random walk.

3 Simulation study

In this section, first we study a random walk to validate our theoretical result namely $\text{Cov Ratio} = 1$. In the second part, we analyze the Constant Elasticity of Variance (CEV) specification with the goal that it will help us interpret any deviation of Cov Ratio from 1. In this part, we explain the CEV specification and give a brief review of the important literature on the CEV specification. We also conduct a simulation exercise to see if the CEV specification is able to capture the level effect as seen later empirically.

3.1 Simulation study for random walk

First, in order to verify the theoretical result that the proposed estimator is unbiased (*Please refer Eqn 7*), we undertake a simulation study for a pure random walk (RW) process following the normal distribution. We generate 10000 paths with $N=10, 20, 30$,

40, 50, 100 steps in the RW model over multiple time windows $k=1, 2, 3, 6, 12, 24, 36, 48$ and 120 months. Across the 10000 paths and for each time window, we computed and stored the values of the mean and standard error of the the estimator *Cov Ratio*. We also conducted *t tests* to test the hypothesis if $E(Cov Ratio) = 1$. The t-stat is computed as $t = \frac{mean-1}{s.e}$. The results are presented in Table 1:

Table 1: Simulation results for *Cov Ratio* for pure random walk following the normal distribution

<i>E(Cov Ratio)</i>										
N/k	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	k=120
N=10	1.001	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
N=20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
N=30	1.001	1.000	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
N=40	1.000	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
N=50	0.999	0.999	1.000	0.999	1.000	1.000	1.000	1.000	1.000	1.000
N=100	1.000	1.001	1.001	1.000	1.000	1.000	1.000	1.000	1.000	1.000
<i>Standard error</i>										
N/k	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	k=120
N=10	7.82E-04	5.04E-04	3.99E-04	2.71E-04	1.89E-04	1.35E-04	1.10E-04	9.48E-05	8.48E-05	5.93E-05
N=20	8.19E-04	5.38E-04	4.30E-04	2.94E-04	2.06E-04	1.45E-04	1.20E-04	1.02E-04	9.19E-05	6.42E-05
N=30	8.43E-04	5.59E-04	4.38E-04	3.04E-04	2.12E-04	1.49E-04	1.22E-04	1.04E-04	9.39E-05	6.56E-05
N=40	8.44E-04	5.58E-04	4.46E-04	3.08E-04	2.17E-04	1.52E-04	1.23E-04	1.08E-04	9.59E-05	6.67E-05
N=50	8.60E-04	5.67E-04	4.51E-04	3.11E-04	2.17E-04	1.54E-04	1.25E-04	1.08E-04	9.52E-05	6.68E-05
N=100	8.69E-04	5.75E-04	4.60E-04	3.15E-04	2.20E-04	1.54E-04	1.26E-04	1.09E-04	9.72E-05	6.73E-05
<i>t – statistics</i>										
N/k	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	k=120
N=10	1.215	-0.993	0.103	-0.089	-0.048	0.080	-0.014	-0.023	0.059	0.036
N=20	0.133	-0.006	-0.017	-0.076	-0.215	0.168	0.125	0.148	0.229	0.145
N=30	1.426	-0.510	-0.596	-0.185	-0.489	-0.217	-0.011	0.057	0.128	0.045
N=40	-0.090	-0.605	-0.467	0.186	0.026	0.058	-0.082	0.020	0.044	-0.087
N=50	-1.699	-1.161	0.370	-0.841	-0.221	-0.114	-0.252	-0.008	0.197	-0.212
N=100	0.001	0.866	0.806	0.088	-0.437	0.216	0.316	-0.175	0.113	0.018

From Table 1, we can see that for all values of N, the mean of the estimator is equal to 1 with very small sampling variance across all k months as validated by the t-statistics. None of the t-stats is significant at 95% confidence level thereby confirming the theoretical result that for a pure random walk process, the proposed estimator is unbiased.

In order to generalize the above findings to any distribution, we ran simulations for a pure random walk process following an asymmetric double exponential distribution. This distribution has the probability density function defined as:

$$f(x) = \begin{cases} p\beta_+e^{-\beta_+|x|} & : x > 0 \\ q\beta_-e^{-\beta_-|x|} & : x \leq 0 \end{cases}$$

where p is the probability; $q = 1 - p$; $\beta > 0$

It can be shown that the probability $p = \frac{\beta_+}{\beta_+ + \beta_-}$. Since we have the constraint that $\frac{1}{\beta_+} * \frac{1}{\beta_-} = \frac{1}{2}$ (Spitzer (1960)), we have set $p = 0.667$.

We generate 10000 paths with N=10, 20, 30, 40, 50, 100 steps in the RW model over the full sample period. Across all values of N, we computed and stored the values of the mean and standard error of the estimator *Cov Ratio*. A t-test was conducted to test the hypothesis if $E(Cov Ratio) = 1$. The t-stat is computed as $t = \frac{mean-1}{s.e}$. The result is presented in Table 2.

From Table 2, we can see that the mean is equal to 1 with very small standard errors for all values of N. None of the t-stats is significant at 95% confidence level across any value of N, thus indicating support for the unbiasedness of the new covariance estimator for this distribution also.

Table 2: Simulation results for the random walk following the asymmetric double exponential distribution

<i>Simulation results for unbiasedness of Cov Ratio</i>				
N	<i>E(Cov Ratio)</i>	<i>Standard error</i>	<i>t – statistics</i>	
10	1.000	0.00E+00	-0.329	
20	1.000	0.00E+00	0.163	
30	1.000	0.00E+00	0.985	
40	1.000	0.00E+00	-1.350	
50	1.000	0.00E+00	-1.030	
100	1.000	0.00E+00	-0.737	

Thus, from Tables 1 and 2, it looks like the proposed estimator is unbiased for a pure random walk following any distribution.

3.2 The Constant Elasticity of Variance (CEV) specification

Now, we introduce the CEV specification because it helps us understand our empirical findings in terms of *Cov Ratio* > 1. (*Please refer to Section 4 for the empirical findings*).

In the classic geometric Brownian motion (GBM), the instantaneous return is given by:

$$\frac{ds}{s} = \mu dt + \sigma dB_t \quad (10)$$

wherein μ is the drift, σ is the volatility, s is price and B_t is a Brownian motion.

For simplicity, let $\mu = 0$; then

$$\frac{ds}{s} = \sigma dB_t \quad (11)$$

The instantaneous variance or quadratic variation (QV) (*on squaring Eqn (11)*) becomes: $E \left[\frac{ds}{s} \right]^2 = \sigma^2 dt$. There is absolutely no level dependence in stock prices volatility in the case of the GBM.

The CEV specification developed by Cox (1996) has been popular in the option pricing literature. CEV model is the simplest way to describe such an inverse relationship as below:

$$ds = \mu s dt + \sigma s^{\frac{\beta}{2}} dB_t \quad (12)$$

Where: μ is the mean return and σ is volatility with $s(t)$ as stochastic price process and B_t is the Brownian Motion. Here, β is the elasticity parameter that defines the relationship between share price level and the variance of the instantaneous rate of return on the share.

The diffusion process in the CEV model when $\mu = 0$ is given by:

$$ds = \sigma s^{\frac{\beta}{2}} dB_t \quad (13)$$

If $\beta = 2$, prices are log-normally distributed and the variance of returns is constant. This is the same as the well-known Black Scholes model. If $\beta < 2$, the stock price is inversely related to the volatility leading to fat left tails. When $\beta > 2$, volatility and the stock price follow the same direction.

The Quadratic variation (*on squaring Eqn (13)*) is given by:

$$d \langle s, s \rangle = \sigma^2 s^\beta dt \quad (14)$$

Cox (1996) restricted this model to: $0 < \beta \leq 2$.

3.3 CEV Model under study

The original CEV model by Cox was restricted to $0 < \beta \leq 2$. However, for the purpose of our study, we extend the model to the case where $\beta > 2$, i.e., we allow for level dependence in stock price volatility in which there is more than proportional increase.

In our study, we are proposing to study the following modified model of CEV specification. Assuming $\mu = 0$, the diffusion process is given by:

$$ds = \sigma s^{1+\delta} dB_t \quad (15)$$

wherein we keep the parameter delta (δ) to lie within the range of $0 < \delta \leq \frac{1}{2}$. Parameter δ controls the relationship between volatility and price, with $\delta = \left(\frac{\beta}{2}\right) - 1$.

Here, the instantaneous return is given by:

$$\frac{ds}{s} = \sigma s^\delta dB_t \quad (16)$$

The Quadratic Variation (*on squaring Eqn 16 and Eqn 17*) are:

$$d \langle s, s \rangle = \sigma^2 s^{2+2\delta} dt \quad (17)$$

$$E \left[\frac{ds}{s} \right]^2 = \sigma^2 s^{2\delta} dt \quad (18)$$

This shows a clear existence of level dependence in stock prices and volatility. The instantaneous returns and volatility depends on the stock price level. Thus, for *Cov Ratio* to be significantly upward biased, we need the parameter delta (δ) to lie within the range of $0 < \delta \leq \frac{1}{2}$.

Remark: If $\delta = 0$, then Eqn 17 becomes:

$$\frac{ds}{s} = \sigma dB_t \quad (19)$$

We can see that this diffusion process is the log normal GBM.

Thus, we can clearly see the existence of level effects in the CEV specification and henceforth we use this specification to explain the bias in the data.

3.4 Literature review on CEV

In this section, we describe below some of the major contributions in the area of CEV.

In a study by [Emanuel and MacBeth \(1982\)](#), they extended their analysis to the case where $\beta > 2$ and discussed its properties. Using their estimation technique, four of the six stocks in their sample were found to have β estimates greater than 2, lending support to the alternative model that they had derived. They found that the implied volatility of

an at-the-money option can also be used to estimate the parameters of the CEV model because it is a reasonable approximation of the true volatility on any day regardless of the value of β . Thus, any value of β can be used with the current stock price, i.e., β varies over time.

[Randall \(1988\)](#) tested the case where $\beta > 2$ under risk neutrality and derived option pricing formula for the same ($-2 \leq \beta \leq 6$). This paper also focused on the estimation of β , with an aim of identifying whether or not the CEV model seems empirically appropriate. He found evidence of results consistent when β takes any value without restriction ($\beta > 0$).

[Schroder \(1989\)](#) and [Dyrting \(2004\)](#) proved that the CEV option pricing formula can be expressed in terms of the non-central chi-square distribution functions. They developed the CEV call option pricing formula in terms of the non-central chi-square distribution for cases when $\beta < 2$ and $\beta > 2$ by treating them separately.

However, [Jackwerth and Rubinstein \(2001\)](#) found that typical values of the β can fit into the market option prices well for post-crash period only when $\beta < 0$ and they called the model with $\beta < 0$ as the unrestricted CEV. In their empirical study, the difference of pricing performance of restricted CEV model ($\beta \geq 0$) and Black Scholes model is not significant.

[Beckers \(1980\)](#) and [MacBeth and Merville \(1980\)](#) concluded that the CEV model works best when β is restricted to be less than 2 ($0 \leq \beta < 2$). However, MacBeth and Merville added that there was evidence in their sample that β changes over time, and that it can be greater than 2.

[Marsh and Rosenfeld \(1983\)](#) in their paper do not model share prices but rather interest rates, using the CEV model and found that the lognormal ($\beta = 2$) model gave the highest likelihood for each of the time series.

[Tucker et al. \(1988\)](#) used the CEV model to describe yet another underlying asset - foreign exchange rates and there was evidence to suggest that β was changing over time for the exchange rates.

[Melino and Turnbull \(1991\)](#) also used the CEV model to describe the evolution of

exchange rates over time and the special case of log normality ($\beta = 2$) was rejected.

To summarize, in much of the existing empirical option pricing literature, β was mostly found to be less than 2 but sometimes it was found to be more than 2. Also, the theoretical option pricing literature has come up with ways of analyzing option prices by developing option pricing models under the CEV specification. Our interest lies in investigating the CEV specification under a more generic pricing framework so as to find an explanation for the case when the newly proposed covariance estimator can be upward biased.

3.5 Simulation study for CEV

The second set of simulation results make use of the CEV specification for studying the biasedness of the proposed estimator. The results indicate that the new estimator is upward biased, validating our argument that this model specification aptly captures the level effects which was not earlier captured by the random walk.

For each delta (δ) = [0, 0.1, 0.2, 0.3, 0.4 and 0.5] fixed over $k=1$ and 120 months respectively, we generated 10000 paths with $N=10, 20, 30, 40, 50, 100$ steps in the CEV process and for each N , we computed and stored the values of mean and standard error of estimator *Cov Ratio*. In order to test the hypothesis if $E(Cov Ratio) = 1$, we conducted t-tests. The t-stat is computed as $t = \frac{mean-1}{s.e}$. The results are presented through Tables 3 and 4 for each of the above k month.

Tables 3 and 4 are divided into 3 sections each. The first section gives the mean of the estimator; the second section gives the standard error and the third section gives the t-statistic. When delta is fixed at 0, CEV model is just the geometric Brownian motion, thus making the *Cov Ratio* unbiased because there is no level effect (*Please refer Eqn 19*). Thus we expect the t-stats to be insignificant for $\delta = 0$ alone.

Table 3 presents the results for $k = 1$ month.

Table 3: Simulation Results for *Cov Ratio* under the CEV Specification for k=1 month

<i>E(Cov Ratio)</i>						
N\δ	δ=0	δ=0.1	δ=0.2	δ=0.3	δ=0.4	δ=0.5
N=10	1.001	1.004	1.010	1.015	1.027	1.023
N=20	1.001	1.004	1.012	1.025	1.040	1.055
N=30	1.000	1.003	1.013	1.028	1.050	1.068
N=40	1.001	1.004	1.015	1.027	1.050	1.074
N=50	1.001	1.004	1.013	1.031	1.056	1.078
N=100	1.000	1.004	1.015	1.034	1.061	1.088
<i>Standard error</i>						
N\δ	δ=0	δ=0.1	δ=0.2	δ=0.3	δ=0.4	δ=0.5
N=10	0.001	0.001	0.001	0.001	0.001	0.002
N=20	0.001	0.001	0.001	0.001	0.002	0.002
N=30	0.001	0.001	0.001	0.001	0.002	0.002
N=40	0.001	0.001	0.001	0.001	0.002	0.002
N=50	0.001	0.001	0.001	0.001	0.002	0.002
N=100	0.001	0.001	0.001	0.001	0.002	0.002
<i>t – statistics</i>						
N\δ	δ=0	δ=0.1	δ=0.2	δ=0.3	δ=0.4	δ=0.5
N=10	0.793	4.175	10.096	12.515	19.352	13.280
N=20	0.951	4.396	11.429	19.439	25.256	29.159
N=30	0.510	3.478	11.780	21.504	31.276	35.202
N=40	0.645	4.183	12.901	20.617	31.005	37.541
N=50	0.557	4.042	11.651	23.367	34.054	39.544
N=100	-0.078	4.190	12.548	24.569	36.443	43.508

From the Table 3 for $k = 1$ month, we can see that for all values of N, mean of the estimator is equal to 1 for $\delta = 0$ as expected and steadily increasing with very small sampling variance as δ and N increases. All the t-stats are significant at 95% confidence level except for when $\delta = 0$. This supports our claim that the proposed estimator is

upward biased under the CEV specification.

Table 4 presents the results for when $k = 120$ months. We can see that for all values of N , when $\delta = 0$, mean of the estimator is equal to 1 as expected. Further, the mean steadily increases as δ and N increases. This is further evidenced by the t-stats. All the t-stats are highly significant at 95% confidence level except for when $\delta = 0$.

Table 4: Simulation Results for <i>Cov Ratio</i> under the CEV Specification for k=120 months						
<i>E(Cov Ratio)</i>						
$N \backslash \delta$	$\delta=0$	$\delta=0.1$	$\delta=0.2$	$\delta=0.3$	$\delta=0.4$	$\delta=0.5$
N=10	1.000	1.003	1.009	1.017	1.022	1.003
N=20	1.000	1.004	1.011	1.024	1.039	1.038
N=30	1.000	1.004	1.013	1.027	1.046	1.053
N=40	1.000	1.004	1.014	1.030	1.051	1.062
N=50	1.000	1.004	1.014	1.031	1.053	1.068
N=100	1.000	1.004	1.015	1.034	1.060	1.082
<i>Standard error</i>						
$N \backslash \delta$	$\delta=0$	$\delta=0.1$	$\delta=0.2$	$\delta=0.3$	$\delta=0.4$	$\delta=0.5$
N=10	1.00E-04	1.00E-04	1.00E-04	1.00E-04	1.00E-04	3.00E-04
N=20	1.00E-04	1.00E-04	1.00E-04	1.00E-04	1.00E-04	3.00E-04
N=30	1.00E-04	1.00E-04	1.00E-04	1.00E-04	2.00E-04	3.00E-04
N=40	1.00E-04	1.00E-04	1.00E-04	1.00E-04	1.00E-04	3.00E-04
N=50	1.00E-04	1.00E-04	1.00E-04	1.00E-04	1.00E-04	3.00E-04
N=100	1.00E-04	1.00E-04	1.00E-04	1.00E-04	1.00E-04	2.00E-04
<i>t – statistics</i>						
$N \backslash \delta$	$\delta=0$	$\delta=0.1$	$\delta=0.2$	$\delta=0.3$	$\delta=0.4$	$\delta=0.5$
N=10	-0.302	45.581	109.530	177.563	159.310	11.389
N=20	0.772	47.721	133.884	231.872	260.990	122.040
N=30	-0.036	50.498	142.241	250.600	298.161	172.192
N=40	-0.479	48.333	151.323	271.253	342.043	214.883
N=50	0.710	49.556	152.974	279.988	360.088	243.718
N=100	-0.863	49.649	164.019	302.012	405.860	329.606

To summarize, from Tables 3 and 4, we can infer that Cov Ratio will be significantly

higher than 1, except for when delta is equal to 0, as seen later in the data too. As delta (δ) increases, the bias in *Cov Ratio* also increases. This can be attributed to the existence of level dependence which is true for all k month periods. These results provide evidence that under the CEV specification, the proposed estimator is upward biased, i.e., *Cov Ratio* is significantly higher than 1.¹

4 Empirical Study

In this section, we analyze empirically a data set of stock indices for four countries: NIFTY (India), S&P 500 (USA), FTSE 100 (UK) and DAX (Germany). Four prices namely the daily opening, high, low and closing prices were taken for the sample period 1st January 1996 up to 30th March 2015, totaling a period of 231 months. The data was downloaded from Bloomberg database.

We have undertaken a rigorous test for testing the unbiasedness of the proposed new covariance estimator *Cov Ratio* by bootstrapping the indices over fixed k month window for the entire sample period with 1000 replications wherein we keep the parameters constant for the duration of the fixed k month interval under study and then allow it to vary across different k months.

We use t-test to determine if *Cov Ratio* = 1 or not. The results are presented in Tables 5-8. Table 5-8 present results for each index respectively for each k month interval individually. In Tables 5-8, we estimate k-month mean and standard error of the estimator for each index. We also conducted a t-test to see if *Cov Ratio* = 1.

The empirical findings based on the estimator *Cov Ratio* are presented in Table 5 for the Nifty Index over all k months. The mean is reasonably stable across k-months, perhaps slightly increasing with respect to k-months. The mean is significantly more than 1 with negligible standard errors as validated by the t-stats. We can also see that across

¹We have also simulated the results for the estimator by fixing delta 0.5, 0.4, 0.3, 0.2, 0.1 and 0 respectively across k= 1, 2, 3, 6, 12, 24, 36, 48, 60 and 120 month time windows with N=[10, 20, 30, 40, 50 and 100] steps across 10000 paths. In all the cases, the mean of the estimator increases as N increases over all k months. The t-stats are significant at 95% confidence level except when delta is fixed at 0, indicating that *Cov Ratio* is significantly higher from 1 and suggesting existence of level effects under the CEV specification.

all k months, t-stat for $Cov\ Ratio = 1$ is highly significant at 95% confidence level leading us to infer that $Cov\ Ratio$ is significantly higher than 1 across all k months and hence is upward biased for the Nifty index.

Table 6 presents the findings for the S&P 500 index. From the table, we can find that $Cov\ Ratio$ is stable and significantly equal to 1 over the sample period with very negligible sampling variance. There is k month effect on bootstrapping beyond 12 months. The t-stat at 95% confidence level is significant up to $k \leq 12$ months. These findings indicate that the estimator is upward biased for the S&P500 up to 12 months.

Table 7 presents the empirical findings for the FTSE 100 index .We can see that mean of the estimator is stable across k months and it is significantly higher than 1. There is no k month effect on bootstrapping. The sample variance is very negligible. The t-stats show that $Cov\ Ratio$ is upward biased for FTSE100 as all the t-stats across the period are highly significant at 95% confidence level.

Table 8 presents the findings for the DAX index over all k months period. We can see that mean of the estimator is stable over the k month interval proving that there is no k month effect on bootstrapping. The mean is significantly equal to 1. The sampling variance is very negligible. The t-stats at 95% confidence level are highly significant over individual k months but not significant in the full sample. This leads us to conclude that $Cov\ Ratio$ is upward biased for DAX index.

Table 5: *Cov Ratio* for the Nifty Index for k-month time windows

For the NIFTY Index: From Jan 1996 to March 2015 (231 months)

	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	Full sample
Number of k-months	231	230	229	226	220	208	196	184	172	
Mean of k-month Cov Ratio	1.05	1.06	1.06	1.07	1.07	1.07	1.07	1.08	1.08	1.08
Stdev of boot mean(k-month Cov Ratio)	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
t-stat for $H_{\{0\}}$: Actual mean of Cov Ratio = 1	4.41	5.20	5.51	5.97	5.70	5.82	5.88	6.20	5.89	6.11

Table 6: *Cov Ratio* for the S&P 500 index k-month time windows

For the S&P 500 Index: From Jan 1996 to Mar 2015 (231 months)										
	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	Full sample
Number of k-months	231	230	229	226	220	208	196	184	172	
Mean of k-month Cov Ratio	1.03	1.03	1.03	1.02	1.02	1.01	1.01	1.01	1.00	1.00
Stdev of mean(k-month Cov Ratio)	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
t-stat for $H_{\{0\}}$: Actual mean of Cov Ratio = 1	3.14	3.11	3.08	2.53	2.02	1.43	0.91	0.56	0.16	0.35

Table 7: *Cov Ratio* for the FTSE 100 index for k-month time windows

For the FTSE 100 Index: From Jan 1996 to Mar 2015 (231 months)											
	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	Full sample	
Number of k-months	231	230	229	226	220	208	196	184	172		
Mean of k-month Cov Ratio	1.09	1.08	1.08	1.08	1.08	1.08	1.07	1.07	1.06	1.07	
Stdev of bootmean(k-month Cov Ratio)	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	
t-stat for $H_{\{0\}}$: Actual mean of Cov Ratio = 1	8.98	8.84	8.82	8.87	9.10	8.71	8.13	6.93	6.58	8.27	

Table 8: *Cov Ratio* for the DAX index for k-month time windows

For the DAX Index: From Jan 1996 to Mar 2015 (231 months)										
	k=1	k=2	k=3	k=6	k=12	k=24	k=36	k=48	k=60	Full sample
Number of k-months	231	230	229	226	220	208	196	184	172	
Mean of k-month Cov Ratio	1.03	1.03	1.03	1.03	1.03	1.03	1.02	1.02	1.02	1.01
Stdev of mean(k-month Cov Ratio)	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
t-stat for $H_{\{0\}}$: Actual mean of Cov Ratio = 1	3.85	3.30	3.25	3.38	3.45	3.08	2.68	2.28	2.10	1.68

Thus, to sum up the findings of the empirical section, we can infer that there is ample evidence against the null hypothesis $Cov\ Ratio = 1$ for all the indices in general. We interpret this as a clear overreaction effect in stock returns. While there is a significant upward bias in the case of NIFTY and FTSE100, the estimator is upward biased for S&P 500 up to $k \leq 12$ months and in the DAX index except for the full sample. We can attribute this upward bias in the estimator to level dependence existing in stock prices, which is aptly captured by the CEV specification as seen in the simulation analysis. (*Please refer to Section 3.5*)

5 Conclusion

In this paper, we have proposed a new modified covariance estimator based on the information contained in daily high, low, opening and closing prices, namely the *Cov Ratio*. This is an improvement over the correlation estimator developed by Rogers and Zhou (2008) as we assume only random walk behavior for the underlying process and not Brownian motion, thus making the estimator very generic. We have proved theoretically that the proposed estimator is unbiased for a random walk. We further validated the theoretical result with the help of simulations. Simulation results suggest that the estimator is unbiased for a pure random walk following any distribution. Empirical analysis provides ample evidence that the estimator is upward biased in general for all the indices under study. That is to say, the estimator is significantly upward biased for NIFTY and FTSE100, while it is upward biased for S&P500 up to $k \leq 12$ months and DAX, except for the full sample.

Since we find that the estimator is upward biased for all the indices, we explore an alternative to random walk under which such a bias can arise. We conjecture this bias as level dependence in stock prices. To capture this effect, we introduce Constant Elasticity of Variance (CEV) specification. With the help of a modified CEV specification for the study, by conducting simulations, we see that it is possible to isolate the effect of intraday level dependence in stock prices using our *Cov Ratio* estimator under the CEV

specification.

Our estimator and its corresponding specification exercise aptly capture the level dependence in stock prices. Our findings will help in risk management by allowing hedgers and traders to take suitable position in derivative contracts and also be of use in portfolio management.

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